ABHINAV SINGH

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EDUCATION

Vellore Institute of Technology, Chennai 2024 – 2028

B.Tech in Computer Science & Engineering (AI & Robotics) | CGPA: 8.75

St. Joseph Convent School (ISC) 2023

Class XII – PCM | 87.0%

St. Joseph Convent School (ICSE) 2021

Class X | 93.4%

TECHNICAL SKILLS

Languages: Python, Java, C, C++

Web & Frameworks: HTML5, CSS3, JavaScript (ES6+), React, Node.js, Flask

AI / ML: Machine Learning (Gradient Boosting, Classification), NLP basics

Embedded & IoT: ESP32, Arduino, C++, ESP-NOW, WebSockets, Sensor Integration

Databases: SQL Plus, SQLite3

Tools & Platforms: Git, GitHub, Linux, Docker (familiar), Scapy

Core CS: Data Structures & Algorithms, OOP, Computer Networks, OS concepts

PROJECTS

Intelligent Firewall & Network Security Protocol Analyzer | *Python, Flask, Scapy, ML* 2026

Computer Networks Course Project

- Built a real-time Intrusion Detection & Prevention System (IDPS) that monitors live network traffic using Scapy, detecting SYN Flood, Port Scan, ICMP Flood, UDP Flood, and Brute Force attacks.
- Integrated a Gradient Boosting ML model to classify traffic as Normal, Suspicious, or Malicious with adaptive threshold tuning, improving detection accuracy under varying load conditions.
- Designed automated IP-blocking with timed auto-unblock, and built a live Flask dashboard visualising alerts, blocked IPs, traffic stats, and protocol distribution in real time.

MineGuard Pro – Smart Safety Helmet System | *ESP32, Arduino, C++, ESP-NOW, WebSockets, HTML/CSS/JS* 2026

- Engineered a real-time underground worker safety system with dual ESP32s communicating over ESP-NOW mesh (router-free), achieving 1-second sensor telemetry and extended multi-zone coverage with zero added infrastructure.
- Designed a priority-ranked C++ alert state machine (SOS > Fall > Gas > Temp > Sound > Heart) processing multi-sensor data in real time, guaranteeing zero missed critical alerts with per-alert buzzer patterns and a physical ACK rescue button.
- Delivered a self-hosted supervisor dashboard served from ESP32 PROGMEM flash, featuring live WebSocket data push, real-time Chart.js trend graphs across 5 sensor channels, and a 20-event incident log — fully offline via device Wi-Fi hotspot.

HACKATHONS & COMPETITIONS

Vehicle Health Monitoring System

- Built a real-time vehicle diagnostics system as part of a team hackathon, capturing engine metrics, fault codes, and sensor data to predict maintenance needs and alert drivers proactively.
- Integrated hardware sensors with a web dashboard to visualise live vehicle health data; secured 5th place out of competing teams.

CivicIssueTracker – Citizen Complaint & Tracking Platform

HTML, CSS, JavaScript, Geotagging APIs **5th Place**

- Co-developed a citizen-driven complaint platform enabling users to report civic issues (water leakage, garbage, sewage) with geotagged locations, built during a hackathon.
- Implemented real-time issue tracking with geospatial mapping so residents and authorities can monitor complaint status and resolution progress on an interactive map.

PROBLEM SOLVING & COMPETITIVE PROGRAMMING

LeetCode — 200+ problems solved across Arrays, Trees, Graphs, DP, and Greedy categories.

HackerRank — Gold Badge in Python · Silver Badge in C.

RELEVANT COURSEWORK

Data Structures & Algorithms · Design & Analysis of Algorithms · Object-Oriented Programming · Computer Networks · Database Management Systems · Operating Systems · Software Engineering · Artificial Intelligence